



Local Recurrence Following Complete Surgical Excision of Primary Merkel Cell Carcinoma

Fergal G. Kavanagh, MB BCh, PhD¹, Hannah L. Kalvin, MSPH², Edmund K. Bartlett, MD¹, Christopher A. Barker, MD³, Charlotte E. Ariyan, MD¹, Mary S. Brady, MD¹, Danielle M. Bello, MD¹, Klaus J. Busam, MD⁴, Katherine S. Panageas, DrPH², and Daniel G. Coit, MD¹

¹Department of Surgery, Memorial Sloan Kettering Cancer Center, New York, NY; ²Department of Epidemiology and Biostatistics, Memorial Sloan Kettering Cancer Center, New York, NY; ³Department of Radiation Oncology, Memorial Sloan Kettering Cancer Center, New York, NY; ⁴Department of Pathology and Laboratory Medicine, Memorial Sloan Kettering Cancer Center, New York, NY

ABSTRACT

Introduction. Merkel cell carcinoma (MCC) is an uncommon neuroendocrine cutaneous malignancy considered to have a high risk of local recurrence (LR). We investigated the true risk of LR in patients with MCC treated by surgical excision alone.

Patients and Methods. Patients with clinically localized MCC excised with pathologically negative surgical margins were identified from a prospectively maintained single-institution database from November 1980 through December 2022. The cumulative incidence of LR, defined as recurrence within 5 cm of the surgical scar, was estimated using death as a competing event.

Results. Among 447 patients, 393 (88%) were treated with surgery alone, and 54 (12%) also received adjuvant radiotherapy (RT) to the primary site. LR occurred in eight patients, with seven recurrences detected within 12 months of treatment. All recurrences occurred in patients treated with surgery alone for a 1-year LR rate of 1.8% (95% confidence interval [CI] 0.8–3.6%). At a median follow-up of 8.8 years (95% CI 7.6–10) from presentation, of the six patients with LR alone, all treated with surgery and/or RT, four patients had no evidence of disease and two died of other causes. Two patients developed distant metastases at the time of LR; both received systemic therapy and died of

disease. The low incidence of LR precluded analysis of risk factors.

Conclusions. Negative-margin surgery without adjuvant RT provides excellent local control, with an LR rate of 1.8%. LR was effectively treated with surgery and/or RT. These data do not support the routine use of adjuvant RT for all patients after complete excision of primary MCC.

Merkel cell carcinoma (MCC) is an uncommon cutaneous neuroendocrine neoplasm. Originally described as “trabecular carcinoma of the skin” by Cyril Toker in 1972,^{1,2} MCC has a rising incidence, projected to surpass 5000 cases annually in the USA by 2030.^{3,4} Nonetheless, the natural history of MCC remains incompletely characterized, with significant uncertainty about the true incidence of local recurrence (LR) after complete surgical excision. Understanding patterns of recurrence is critical, as treatment decisions often rely on historical data that may not reflect contemporary outcomes.

Early studies reported LR rates of 30–39%, shaping the view of MCC as an aggressive disease. These estimates, presented in Table 1, are now under scrutiny, with modern data increasingly challenging this view.^{5–22} Recent single-institution studies using stricter definitions report lower LR rates. Earlier studies often had inconsistent definitions conflating regional and in-transit disease present at diagnosis with LR, artificially inflating concerns about MCC’s local behavior and overstating the need for adjuvant RT. Current National Comprehensive Cancer Network (NCCN) guidelines endorse RT for most patients beyond the lowest-risk category, even after complete resection with negative

TABLE 1 Reported local recurrence rates in MCC across historical series by treatment modality

Study	Year	Number of patients	Surgery only	Surgery and RT	Unclear/multi-modality	LR rate
Sibly et al.	1985	43	41	3		30%
O'Brien et al.	1987	11	10	1		36%
Morrison et al.	1990	54	37	17		24%
Shaw et al.	1991	234			234	39%
Yiengpruksawan et al.	1991	70				25%
O'Connor et al.	1997	86	50	7		32%
Allen et al.	1999	102	88	14		8%
Gollard et al.	2000	22	16	6		4.5%
Snow et al.	2001	12	9	3		22%
Boyer et al.	2002	45	25	20		16%*
Brissett et al.	2002	22	22			18%
Koljonen et al.	2003	34	17	12	5	29%
Hohaus et al.	2003	16	4	12		19%
Veness et al.	2005	86	37	25	24	14%
Jouary et al.	2012	83	44	39		4.5%
Soult et al.	2012	26	13	13		15%
Frohm et al.	2016	104	104			1%
McEvoy et al.	2022	618			618	2.75%

*Three of the four LRs were in-transit metastasis

In the majority of these publications, there was no timepoint associated with the LR rate

margins, despite a lack of prospective data.²³ Adjuvant RT is recommended for patients with adverse features such as tumor size > 1 cm, lymphovascular invasion (LVI), head and neck location, chronic T cell suppression, history of chronic lymphocytic leukemia (CLL), and current human immunodeficiency virus (HIV) infection.²⁴ These recommendations rely primarily on retrospective administrative data, which are limited by heterogeneous cohorts, use of overall rather than disease-specific survival, incomplete recurrence documentation, and selection bias.^{25–27}

In this context, we present a detailed analysis from the largest single-institution MCC cohort with prospectively maintained data spanning over 40 years. This study provides granular physician-curated data on recurrence patterns, timing, and salvage outcomes, enabling a robust assessment of LR risk and the utility of adjuvant RT. Our findings not only demonstrate a markedly lower LR rate than historically reported but also provide compelling evidence that LR, when it occurs, is a highly salvageable event. These insights challenge prevailing treatment paradigms and raise critical questions about the routine application of adjuvant RT in localized MCC.

PATIENTS AND METHODS

Patient Selection

Patients were identified from a database of MCC cases treated at Memorial Sloan Kettering Cancer Center (MSK) from 1980 through the end of 2022. The database has been prospectively maintained since 2001. Institutional review board approval was obtained for this study.

The current study focused on patients who presented for initial management of clinical stage I–II MCC (per the American Joint Committee on Cancer [AJCC] 8th edition) within 60 days of diagnosis and who subsequently underwent a pathological margin-negative excision of their primary tumor.²⁴ Patients initially treated at other institutions or seen in consultation without receiving treatment at MSK were excluded. Patients who presented with clinical regional nodal disease or with distant metastatic disease were excluded.

Patients whose tumors were excised with positive margins and those treated with therapeutic radiation therapy (RT) alone were not included in the primary analysis.

Patient variables analyzed included age; sex; history of prior, synchronous, or other malignancies (cutaneous, hematologic, or solid tumor); and presence of adverse risk factors as defined by NCCN guidelines such as tumor size > 1 cm, LVI, head or neck location, chronic T cell suppression, history of CLL, and active HIV infection.²⁴ All patients

included in the analysis had histologically confirmed MCC, reviewed by a dedicated dermatopathologist (K.J.B.), as previously published.²⁸ Tumor characteristics included tumor size, location, and LVI (absent, present, or suspicious). Merkel cell polyoma virus (MCPyV) status (as detected on IHC for CM2B4), when available, was reported. Treatment variables included the clinically measured radial skin margin. Both the initial biopsy and subsequent wide excision specimens were considered in pathologic determination of tumor size and margin status.

Adjuvant Radiation Therapy

Adjuvant RT consisted of ≥ 50 Gy of RT administered to the primary tumor site after pathologically confirmed margin-negative wide excision.

Outcome Measures

The primary outcome was any LR, defined as tumor recurrence arising within 5 cm of the primary excision scar, in patients following negative-margin resections. Records of all patients with recurrence were reviewed by a panel of three physicians (D.G.C., E.B., and C.B.) to confirm the date and type of recurrence. Recurrence was identified by the patient, clinician, or imaging, with the date of recurrence recorded as the first documented evidence in the medical record. Patients were recommended to follow on a schedule consistent with good clinical practice, every 3–4 months for the first 2–3 years, then every 6–12 months thereafter.

Statistical Analysis

Descriptive statistics were used to summarize baseline demographic, clinical, and pathological variables. Categorical variables were reported as counts and percentages; continuous variables were summarized as medians with interquartile ranges (IQR). Comparisons between treatment groups (surgery alone versus surgery and RT) and between patients who did or did not develop LR within 1 year of surgery were performed using Wilcoxon rank-sum tests for continuous variables and Pearson's chi-squared or Fisher's exact tests for categorical variables. Analyses were presented with correction for multiple comparisons using the Benjamini–Hochberg method for false discovery rate. Follow-up time was calculated from date of presentation using reverse Kaplan–Meier methodology.

Time-to-event analyses were limited to patients that underwent surgery and calculated from the date of surgical resection to recurrence, death, or last follow-up. Cumulative incidence functions were used to estimate local recurrence (LR), accounting for the competing risks of recurrence in other sites (nodal, regional, or distant) and death without

recurrence. The association of potential risk factors with LR was assessed by logistic regression.

All statistical analyses were conducted by H.K. and K.P. The final analysis was updated on 24 July 2025. Analyses were conducted using R version 4.4.0. All two-sided *p*-values < 0.05 were considered significant.

RESULTS

Patient Selection

Of the 1160 patients included in the MSK database, initial treatment was provided to 721 patients within 60 days of diagnosis (Fig. 1). Of these, 90 patients were excluded on the basis of clinically positive nodes, 31 for distant metastatic disease, and 114 for unknown primary site. Another 24 patients were treated with RT alone to the primary site, and 15 patients had positive surgical margins; these cohorts were excluded from the primary analysis and are described in Supplementary Tables 1 and 2, respectively.

Patient, Tumor, and Treatment Characteristics

A total of 447 patients with clinical stage 1 or 2 disease underwent negative-margin resection (Table 2). The median age at diagnosis was 72 years (IQR 64–78), and 255 (57%) were male. Of the 283 patients with information available, 150 (53%) had prior or synchronous cutaneous malignancy, 27 (9.5%) had prior or synchronous hematological malignancies, and 97 (34%) had prior or synchronous solid tumors.

Other MCC adverse risk factors were observed infrequently: chronic steroid use (> 1 month) was the most prevalent ($n = 17$), followed by chronic lymphocytic leukemia (CLL) ($n = 7$), solid organ transplant ($n = 5$), and HIV infection ($n = 3$).

The median clinical tumor size was 12 mm (IQR 7–20 mm). The most common primary tumor location was the extremity (232 patients, 52%), followed by the head or neck (141, 32%). Of the 308 patients for whom LVI was reported, it was present or suspicious in most (228 [74%] and 18 [5.8%], respectively). In the 62 patients for whom Merkel cell polyomavirus (MCPyV) status was reported, it was positive in 47 (76%) and negative in 15 (24%). The median clinical margin was 15 mm (IQR 10–20); 86% of patients had margins in the 10–20 mm range.

Comparison of Treatment Groups

Of the 447 patients, 393 underwent surgery alone, and 54 received subsequent adjuvant RT to the primary site. Compared with those who underwent surgery only, those

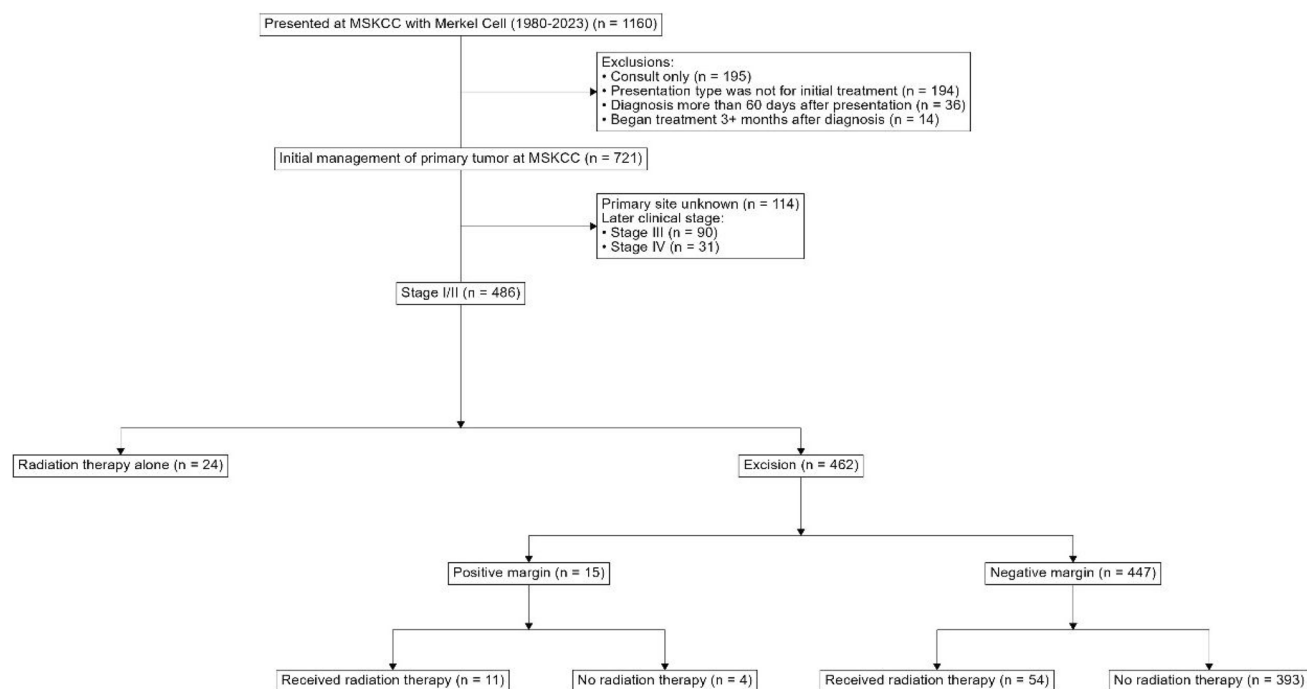


FIG. 1 Enrollment diagram

receiving adjuvant RT were younger (median age 68 years [IQR 62–74] versus 72 years [IQR 65–78], $p = 0.01$) and had larger tumors (median clinical size 20 mm [IQR 11–39] versus 12 mm [IQR 7–20], $p < 0.001$) (Table 2). Adjuvant RT-treated tumors were more likely to display LVI (44/54 [81%] versus 184/393 [47%], for no adjuvant RT ($p < 0.001$) and, counterintuitively, had wider median clinical margins (20 mm [IQR 15–20] versus 15 mm [IQR 10–20] for no adjuvant RT, ($p = 0.002$).

Outcomes: LR

All eight patients who developed LR after margin-negative excision were initially treated with surgery alone. At 12 months, the cumulative incidence of local recurrence was 1.8% (CI 0.8–3.6%) (Fig. 2). All LRs were observed within 14 months of resection. Six patients presented with LR-only recurrence; two patients presented with LR and simultaneous regional and distant metastases, both within 6 months of initial treatment. Table 3 lists the characteristics, management, and outcomes of patients with LR. The median age was 66 years (IQR 57–76). Median clinical tumor size was 14 mm (IQR 7–24). LVI or tumor emboli were present in five patients (63%), absent in two (25%), and unspecified in one (13%). LR occurred in the head or neck in three patients and in the extremities in five. The median clinical surgical margin was 12.5 mm (IQR 10–20) (Table 3).

Management of LR

Treatment of LR included surgery and RT ($n = 3$), RT alone ($n = 2$), and surgery alone ($n = 1$). In the patients with LR and regional and distant metastatic disease, multimodal therapy was used, including chemotherapy and immunotherapy ($n = 2$). Four patients had no evidence of disease (NED) at last follow up, two died of other causes (DOC), and two died of disease (DOD). The two patients who died of disease both developed recurrences early with synchronous regional and distant metastases, suggesting a biologically aggressive phenotype rather than local treatment failure. In every case of isolated LR where LR-directed therapy was employed, long-term durable local control was achieved.

NCCN High-Risk Factors

Among the entire cohort of 447 patients, 345 (77%) had ≥ 1 NCCN MCC adverse risk factor (tumor size > 1 cm, LVI, head or neck location, chronic T cell suppression, current or previous CLL, active HIV infection) that would have led to the recommendation to receive adjuvant RT according to current guidelines.

Factors Associated with Increased Risk of LR

Table 4 presents a comparison of patient, tumor, and treatment-related factors by status of LR at 12 months. The

TABLE 2 Baseline demographic, tumor, and treatment characteristics of patients treated with surgery alone versus surgery and radiation therapy

Characteristic	Overall <i>N</i> = 447	Surgery <i>n</i> = 393	Surgery and RT <i>n</i> = 54	<i>P</i> -value
Age at diagnosis (years)	72 (64–78)	72 (65–78)	68 (62–74)	0.01
Sex				0.3
Female	192 (43%)	172 (44%)	20 (37%)	
Male	255 (57%)	221 (56%)	34 (63%)	
Prior or synchronous cutaneous malignancy				
Cutaneous	150 (33%)	135 (34%)	15 (28%)	0.3
Hematologic	27 (6%)	24 (6%)	3 (5.6%)	> 0.9
Solid tumor	97 (21%)	85 (25%)	12 (22%)	0.9
NCCN high-risk factors	345 (77%)	293 (75%)	52 (96%)	
Clinical tumor diameter (mm)	12 (7–20)	12 (7–20)	20 (11–39)	< 0.001
Tumor size (grouped)				< 0.001
< 10 mm	151 (34%)	142 (36%)	9 (17%)	
10–20 mm	194 (43%)	174 (44%)	20 (37%)	
> 20 mm	102 (23%)	77 (20%)	25 (46%)	
Clinical margin (mm)	15 (10–20)	15 (10–20)	20 (15–20)	0.002
Unknown	44	38	6	
Clinical margin (grouped)				0.7
< 10 mm	25 (6.2%)	22 (6.2%)	3 (6.3%)	
10–20 mm	347 (86%)	307 (86%)	40 (83%)	
> 20 mm	31 (7.7%)	26 (7.3%)	5 (10%)	
Primary site				0.06
Extremity	232 (52%)	201 (51%)	31 (57%)	
Head/neck	141 (32%)	131 (33%)	10 (19%)	
Trunk, buttock, or vulva	74 (17%)	61 (16%)	13 (24%)	
Lymphovascular invasion or tumor emboli				< 0.001
Yes	228 (51%)	184 (47%)	44 (81%)	
Unspecified	139 (31%)	132 (34%)	7 (13%)	
MCPyV status (CM2B4 IHC)				
Positive	47 (11%)			
Negative	15 (3.4%)			
Not reported	385 (86%)			

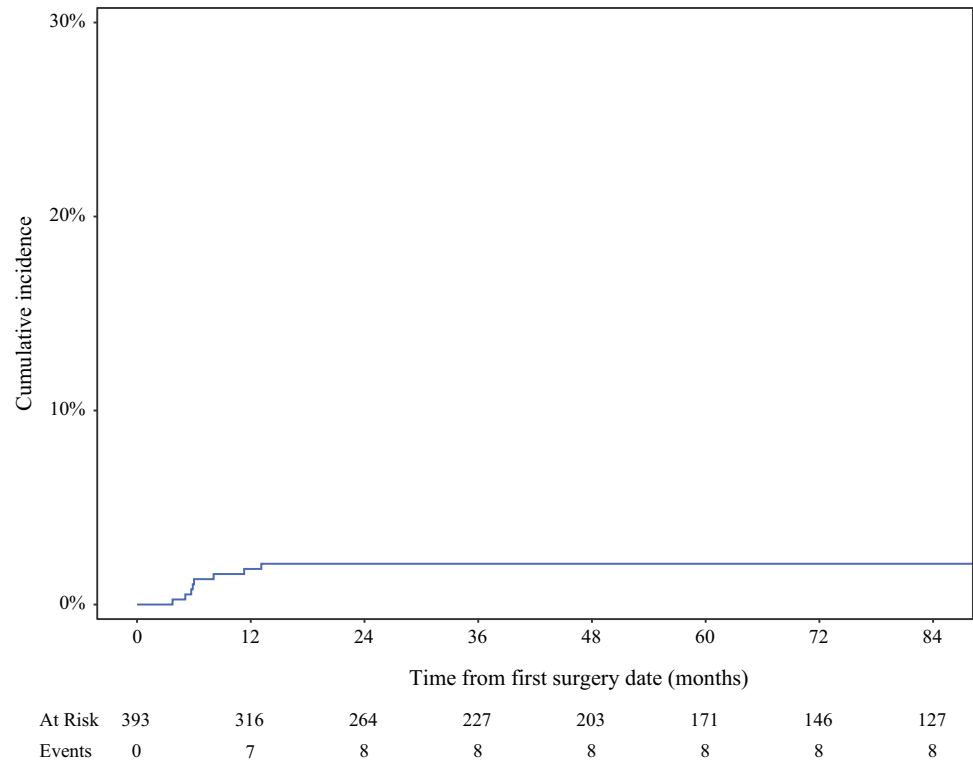
N (%) or median (interquartile range)

lowest *p*-values were observed for chronic steroid use and solid organ transplantation; however, after adjustment for false discovery rate (*q*-value), associations were not statistically significant. Similarly, although NCCN high-risk features were slightly more frequent among patients with LR (6/7; 86% versus 317/411; 77%), this difference was not statistically significant. The low event rate (7 of 419) limited the statistical power of this analysis.

DISCUSSION

The current study represents the largest single-institution cohort of patients with clinically localized MCC undergoing confirmed margin-negative excision. At a median follow-up of 8.8 years, the cumulative incidence of LR in the surgery-alone group was 1.8%. All LRs were seen within 14 months

after initial treatment. All six patients with isolated LR received additional LR-directed therapy, and all achieved durable long-term local control. None of these six patients subsequently died of MCC. In contrast, the two patients who experienced early LR in association with regional and distant metastatic disease both progressed rapidly despite systemic therapy and died of disease, suggesting a much more aggressive biology. Owing to the low event rate, we were unable to identify any factors, including multiple NCCN-defined adverse features, as associated with increased risk of LR. Notably, 77% of the cohort had ≥ 1 NCCN-defined adverse features, yet few developed recurrences. These findings underscore the low incidence of LR after negative-margin excision and the poor discriminatory power of current risk stratification models, raising concerns about potential overtreatment with adjuvant RT.

FIG. 2 Cumulative incidence of LR in surgery-only patients**TABLE 3** Clinical and treatment characteristics of patients with local recurrence

	Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6	Patient 7	Patient 8
Sex	Female	Male	Male	Male	Male	Male	Male	Male
Age (years)	67	65	76	39	76	57	81	55
Tumor size	40 mm	7 mm	4 mm	15 mm	5 mm	12 mm	20 mm	20 mm
Location	Extremity	Head/neck	Head/neck	Extremity	Head/neck	Extremity	Extremity	Extremity
Lymphovascular invasion or tumor emboli	Not recorded	No	Yes	Yes	No	Yes	Yes	Yes
Margin	28 mm	0.5 mm	10 mm	20 mm	10 mm	15 mm	10 mm	20 mm
Interval to recurrence	4 months	6 months	13 months	6 months	11 months	8 months	6 months	5 months
Year of LR*	1995	2000	2011	2012	2020	2019	2019	2023
Regional/distant metastasis	Yes	No	No	No	No	No	Yes	No
Treatment	Chemotherapy	Surgery	Surgery	Surgery	Radiation	Radiation	Immunotherapy	Surgery and radiation
Local control	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Final status	Died of disease	Died of other cause	Died of other cause	No evidence of disease	No evidence of disease	No evidence of disease	No evidence of disease	No evidence of disease

*LR local recurrence

Our definition of LR as recurrence within 5 cm of the scar was somewhat arbitrary; no standardized definition was available from literature. Our definition is based on a similarly arbitrary definition for LR in melanoma: tumor

regrowth within 2 cm of the surgical site following a definite excision with appropriate surgical margins.²⁹ Recurrences that are > 2 cm from the primary lesion but not beyond the regional nodal basin are termed in-transit metastases thought

TABLE 4 Univariable analysis of association of clinicopathologic factors with risk of local recurrence within 12 months

Characteristic	Overall <i>N</i> = 419	No LR in 12 months <i>n</i> = 412	LR within 12 months <i>n</i> = 7	<i>P</i> -value	<i>Q</i> -value ¹
Age at diagnosis, years	71 (64–77)	72 (64–77)	65 (57–72)	0.2	0.5
Sex				0.2	0.6
Female	178 (42%)	177 (43%)	1 (14%)		
Male	241 (58%)	235 (57%)	6 (86%)		
Primary site				0.6	>0.9
Extremity	218 (52%)	213 (52%)	5 (71%)		
Head/neck	130 (31%)	128 (31%)	2 (29%)		
Trunk, buttock, or vulva	71 (17%)	71 (17%)	0 (0%)		
Clinical tumor diameter (mm)	13 (7–20)	13 (7–20)	15 (10–29)	0.4	0.9
Tumor size (grouped)				> 0.9	> 0.9
< 10 mm	142 (34%)	140 (34%)	2 (29%)		
10–20 mm	180 (43%)	177 (43%)	3 (43%)		
> 20 mm	97 (23%)	95 (23%)	2 (29%)		
LVI/tumor emboli				0.2	0.5
Yes	212 (51%)	207 (50%)	5 (71%)		
No	56 (13%)	54 (13%)	2 (29%)		
Suspicious	17 (4.1%)	17 (4.1%)	0 (0%)		
Unspecified	134 (32%)	134 (33%)	0 (0%)		
SLN status				0.2	0.4
Positive	125 (30%)	54 (13%)	2 (29%)		
Negative	190 (45%)	189 (46%)	1 (14%)		
Unknown/not done	104 (25%)	102 (25%)	2 (29%)		
Clinical margin size	15 (10–20)	15 (10–20)	15 (10–20)	0.9	> 0.9
Unknown	39	39	0		
Clinical margin size (grouped)				> 0.9	> 0.9
< 10 mm	23 (6.1%)	23 (6.2%)	0 (0%)		
10–20 mm	327 (86%)	320 (86%)	7 (100%)		
> 20 mm	30 (7.9%)	30 (8.0%)	0 (0%)		
Unknown	39	39	0		
Chronic steroid use (> 1 month)	15 (3.6%)	13 (3.2%)	2 (29%)	0.023	0.3
Chronic lymphocytic leukemia	7 (1.7%)	7 (1.7%)	0 (0%)	> 0.9	> 0.9
Solid organ transplant	3 (0.7%)	2 (0.5%)	1 (14%)	0.049	0.3
HIV infection (with or without AIDS)	2 (0.5%)	2 (0.5%)	0 (0%)	> 0.9	> 0.9
Has ≥ 1 NCCN high-risk factor	323 (77%)	317 (77%)	6 (86%)	> 0.9	> 0.9

Bold values indicate statistical significance $P < 0.05$

¹False discovery rate correction for multiple testing

Limited to patients that had a local recurrence in 12 months from surgery or at least 12 months of follow-up

N (%) or median (interquartile range)

AIDS acquired immunodeficiency syndrome, *SLN* sentinel lymph node status

to be manifestation of intralymphatic tumor dissemination.³⁰ Six of the LRs within our cohort occurred within the surgical scar, while the other two were posterior and distal to the skin graft site, respectively. They occurred within what would have been a reasonable adjuvant RT field and would not have been considered regional or in-transit disease.

Historically, the perception of MCC as an aggressive tumor was shaped by widely disparate reports of LR rates,

ranging from < 1–39%.^{5–19} This variability reflects the limitations of historical data, namely small sample sizes, retrospective designs, and inconsistent definitions. An early and widely cited meta-analysis of 1254 patients across 132 studies reported a 40% local recurrence rate but included 38 case reports, lacked recurrence site differentiation, and averaged less than six patients per study.³¹ While that analysis also claimed that RT reduced LR rates, no details were provided

on RT dose or field. Despite its methodological flaws, that study is often quoted as the rationale for routine adjuvant RT after complete excision of a primary MCC.

MCC is radiosensitive, but high-level evidence supporting adjuvant RT to mitigate LR risk remains lacking. NCCN recommends adjuvant RT for all but the lowest-risk patients, including those with narrow margins or clear margins with adverse features such as tumor size > 1 cm, LVI, head or neck location, and immunosuppression.²⁴ These recommendations are based largely on retrospective administrative datasets with their inherent limitations.^{25,27,32} First, administrative datasets are heterogeneous, with variable tumor and treatment characteristics. Second, overall survival is used as an outcome measure, which is problematic in an elderly population with competing mortality risks; disease-specific survival would be more appropriate but is not available. Third, and relevant to the current study, administrative datasets have no granular and consistent information on site, timing, and subsequent treatment of recurrence.

The only prospective randomized trial of RT in MCC, conducted in France, examined the efficacy of RT to the nodal basin of patients with localized MCC undergoing wide excision without sentinel lymph node biopsy.¹⁶ All patients enrolled received RT to the primary tumor surgical bed. While nodal RT reduced regional recurrence in clinical stage I disease, there was no improvement in progression-free or overall survival. Furthermore, the low LR rates in the current study would suggest analyses of administrative databases that identify timing of adjuvant radiation as important are likely identifying noncausal associations rather than a meaningful impact on survival.²⁶ However, this local RT paradigm has permeated the field: A recent publication of a Seattle-based cohort reported that 92% of patients with MCC received local adjuvant RT following complete excision.³³ Consistent with this, evidence from Australian centers suggest that narrower surgical margins may be appropriate when combined with adjuvant radiation therapy, and that excellent control can also be achieved with radiation therapy alone as the primary treatment for stage I–III disease.^{34,35}

Our data show that LR is rare and salvageable when it occurs. Importantly, when LR manifests concurrently with distant disease, it likely signifies underlying aggressive tumor biology rather than a failure of the initial local treatment modalities. In contrast to large administrative database studies, our data include detailed, nuanced information regarding the specific type and time of recurrence and its management. We demonstrated that most LRs occur within the first year of treatment and can be effectively managed when isolated. Notably, there were no LRs in the group that received RT. But with a 12-month cumulative incidence of only 1.8%, the event rate is so low that the group functions more as a control. To justify radiation, the baseline risk

would need to be even lower—raising doubt about its utility in this setting.

However, the NCCN recommends adjuvant RT to reduce LR in MCC on the basis of adverse risk factors, but the supporting evidence is weak.²³ A key study cited by the NCCN to support adjuvant RT reveals that most initial disease progressions were regional (61.8%) or satellite/in-transit (24.2%), with distant metastases developing in nearly half of all patients (47.8%) over time.³⁶ Furthermore, many studies referenced in the NCCN guidelines disproportionately focus on immunosuppressed populations, who are indeed at an elevated risk of *developing* MCC.^{37,38} While immunosuppression has been inconsistently linked to poorer MCC-specific survival, it does not uniformly predict LR, and the supporting evidence for the guidelines is not convincing.³⁹ These criteria may conflate a patient's risk for developing MCC and/or their risk of systemic relapse with their actual risk of LR. Our study's evaluation of LR predictors, though limited by the rarity of events, found only a possible trend toward association with chronic T cell suppression and CLL. Critically, a majority (77%) of patients without LR in our cohort possessed ≥ 1 NCCN high-risk feature, suggesting that they are not reliable predictors and should not be utilized as criteria for adjuvant RT after margin-negative excision. Notably, 50% of the LRs occurred in primary tumors that were considerably larger (1.5–4 cm), and three of the four remaining LRs were located in the head and neck. While not statistically significant, this pattern appears consistent with one of the NCCN-defined high-risk features.

Six criteria should guide decisions about adjuvant therapy. First, the absolute risk of the event: We observed a 1.8% LR rate, meaning that > 98% of patients could not possibly benefit from adjuvant RT. Second, the absolute risk reduction: No RT-treated patients developed LR, implying a potential 1.8% benefit. Third, treatment costs, including not just financial burden but also physical and financial toxicity, including rare complications such as osteoradionecrosis and post-RT sarcoma (both observed in our series), and the logistical burden of treatment.^{40–42} Fourth, the ability to salvage LR without upfront RT: All isolated LRs were effectively treated with additional local therapy. Fifth and sixth are physician and patient preference, both of which should be informed by accurate data on MCC's natural history. Although most patients in our series were treated with excision alone, a small proportion were selectively offered radiation, likely on the basis of clinical judgment.

This study is derived from the largest single-institution MCC database spanning over 40 years, prospectively maintained for the last 25 years. The strengths of the study include highly granular physician-curated data on patient, tumor, and treatment variables, with particular attention focused on patterns of recurrence and long-term disease-specific outcomes, information completely lacking from

large multicenter or administrative database reviews. The treatments administered were straightforward and widely available. As such, we strongly believe that our observations are not only applicable to patients treated at our institution but should also be broadly generalizable.

All retrospective analyses, even those examining prospectively maintained databases, have inherent limitations. Over the course of the study, we had no consistent formal institutional synoptic pathology reporting, and as such, there are a substantial number of missing observations on the presence of LVI and MCPyV status. Although most clinical surgical margins were 10–20 mm, these were not standardized. In contrast to widespread practice elsewhere, there was an institutional bias that, for most patients, surgery alone was sufficient. Referral criteria for adjuvant RT were not standardized; we could only make inferences by looking retrospectively at the patient, tumor, and treatment characteristics available to us.

In conclusion, this study provides high-quality evidence that the cumulative incidence of LR after margin-negative excision for MCC is low (1.8%) and, in the absence of synchronous metastatic disease, salvageable by additional local therapy. These findings challenge the current paradigm that favors routine adjuvant RT for most patients. Given the low event rate, we were unable to identify reliable predictors of LR. Ultimately, our data do not support the routine use of adjuvant RT for any patient following complete resection of localized MCC.

SUPPLEMENTARY INFORMATION The online version contains supplementary material available at <https://doi.org/10.1245/s10434-025-18670-2>.

DISCLOSURES C.A.B. has served on advisory boards for Regeneron. C.E.A. has consulted or served on advisory boards for Iovance Biotherapeutics, Merck KGaA, and Merck & Co. K.J.B. holds a paid position with the American Society of Dermatopathology. E.B. declares QBiotics Group, Ad Board, and institutional research support; Skyline DX, institutional research support. No other authors have relationships with outside entities to disclose. This research was supported in part through the MSK institutional NIH/NCI Cancer Center Support Grant P30 CA008748, which funds the institution's core resources.

REFERENCES

1. Toker C. Trabecular carcinoma of the skin. *Arch Dermatol.* 1972;105(1):107–10.
2. Toker C. Trabecular carcinoma of the skin: a question of title. *Am J Dermatopathol.* 1982;4(6):497–500.
3. Paulson KG, Park SY, Vandeven NA, et al. Merkel cell carcinoma: current US incidence and projected increases based on changing demographics. *J Am Acad Dermatol.* Mar 2018;78(3):457–463 e452.
4. Jacobs D, Huang H, Olino K, Weiss S, Kluger H, Judson BL, Zhang Y. Assessment of age, period, and birth cohort effects and trends in Merkel cell carcinoma incidence in the United States. *JAMA Dermatol.* 2021;157(1):59–65.
5. Rk S. Primary neuroendocrine (Merkel cell?) carcinoma of the skin 1&2. *Am J Surg Pathol.* 1985;9:95–116.
6. O'Brien P, Denham J, Leong ASY. Merkel cell carcinoma: a review of behaviour patterns and management strategies. *Aust N Z J Surg.* 1987;57(11):847–850.
7. Morrison WH, Peters LJ, Silva EG, Wendt CD, Ang KK, Goepfert H. The essential role of radiation therapy in securing locoregional control of Merkel cell carcinoma. *Int J Radiat Oncol Biol Phys.* 1990;19(3):583–91.
8. Shaw J, Rumball E. Merkel cell tumour: clinical behaviour and treatment. *J Br Surg.* 1991;78(2):138–42.
9. Yiengpruksawan A, Coit DG, Thaler HT, Urmacher C, Knapper WK. Merkel cell carcinoma. Prognosis and management. *Arch Surg.* Dec 1991;126(12):1514–1519.
10. O'Connor WJ, Roenigk RK, Brodland DG. Merkel cell carcinoma: comparison of Mohs micrographic surgery and wide excision in eighty-six patients. *Dermatol Surg.* 1997;23(10):929–33.
11. Allen PJ, Zhang ZF, Coit DG. Surgical management of Merkel cell carcinoma. *Ann Surg.* 1999;229(1):97–105.
12. Gollard R, Weber R, Kosty MP, Greenway HT, Massullo V, Humberson C. Merkel cell carcinoma: review of 22 cases with surgical, pathologic, and therapeutic considerations. *Cancer.* 2000;88(8):1842–51.
13. Snow SN, Larson PO, Hardy S, et al. Merkel cell carcinoma of the skin and mucosa: report of 12 cutaneous cases with 2 cases arising from the nasal mucosa. *Dermatol Surg.* 2001;27(2):165–70.
14. Boyer JD, Zitelli JA, Brodland DG, D'Angelo G. Local control of primary Merkel cell carcinoma: review of 45 cases treated with Mohs micrographic surgery with and without adjuvant radiation. *J Am Acad Dermatol.* 2002;47(6):885–92.
15. Veness MJ, Perera L, McCourt J, Shannon J, Hughes TM, Morgan GJ, Gebbski V. Merkel cell carcinoma: improved outcome with adjuvant radiotherapy. *Aust N Z J Surg.* 2005;75(5):275–81.
16. Jouary T, Leyral C, Dreno B, et al. Adjuvant prophylactic regional radiotherapy versus observation in stage I Merkel cell carcinoma: a multicentric prospective randomized study. *Ann Oncol.* 2012;23(4):1074–80.
17. Frohm ML, Griffith KA, Harms KL, et al. Recurrence and survival in patients with merkel cell carcinoma undergoing surgery without adjuvant radiation therapy to the primary site. *JAMA Dermatol.* 2016;152(9):1001–7.
18. McEvoy AM, Lachance K, Hippe DS, et al. Recurrence and mortality risk of Merkel cell carcinoma by cancer stage and time from diagnosis. *JAMA Dermatol.* 2022;158(4):382–9.
19. Soult MC, Feliberti EC, Silverberg ML, Perry RR. Merkel cell carcinoma: high recurrence rate despite aggressive treatment. *J Surg Res.* 2012;177(1):75–80.
20. Brissett AE, Olsen KD, Kasperbauer JL, et al. Merkel cell carcinoma of the head and neck: a retrospective case series. *Head Neck.* 2002;24(11):982–8.
21. Koljonen V, Böhling T, Granthor G, Tukiainen E. Merkel cell carcinoma: a clinicopathological study of 34 patients. *Eur J Surg Oncol.* 2003;29(7):607–10.
22. Hohaus K, Köstler E, Schönlebe J, Klemm E, Wollina U. Merkel cell carcinoma—A retrospective analysis of 17 cases. *J Eur Acad Dermatol Venereol.* 2003;17(1):20–4.
23. Schmults CD, Blitzblau R, Aasi SZ, et al. NCCN Guidelines® insights: Merkel cell carcinoma, Version 1.2024: featured updates to the NCCN guidelines. *J Natl Compr Cancer Netw.* 2024;22(1D).
24. Amin MB, Greene FL, Edge SB, et al. The Eighth Edition AJCC Cancer Staging Manual: continuing to build a bridge from a

- population-based to a more “personalized” approach to cancer staging. *CA Cancer J Clin*. Mar 2017;67(2):93-99.
25. Bhatia S, Storer BE, Iyer JG, et al. Adjuvant radiation therapy and chemotherapy in Merkel cell carcinoma: survival analyses of 6908 cases from the National Cancer Data Base. *J Natl Cancer Inst*. Sep 2016;108(9).
 26. Alexander NA, Schaub SK, Goff PH, et al. Increased risk of recurrence and disease-specific death following delayed postoperative radiation for Merkel cell carcinoma. *J Am Acad Dermatol*. 2024;90(2):261–8.
 27. Petrelli F, Ghidini A, Torchio M, et al. Adjuvant radiotherapy for Merkel cell carcinoma: a systematic review and meta-analysis. *Radiother Oncol*. 2019;134:211–9.
 28. Andea AA, Coit DG, Amin B, Busam KJ. Merkel cell carcinoma: histologic features and prognosis. *Cancer*. 2008;113(9):2549–58.
 29. Balch CM, Urist MM, Karakousis CP, et al. Efficacy of 2-cm surgical margins for intermediate-thickness melanomas (1 to 4 mm) results of a multi-institutional randomized surgical trial. *Ann Surg*. 1993;218(3):262–9.
 30. Pawlik TM, Ross MI, Thompson JF, Eggermont AM, Gershenwald JE. The risk of in-transit melanoma metastasis depends on tumor biology and not the surgical approach to regional lymph nodes. *J Clin Oncol*. 2005;23(21):4588–90.
 31. Lewis KG, Weinstock MA, Weaver AL, Otley CC. Adjuvant local irradiation for Merkel cell carcinoma. *Arch Dermatol*. 2006;142(6):693–700.
 32. Vayntraub A, Tayeb N, Squires B, et al. The Association of radiation therapy and chemotherapy on overall survival in Merkel cell carcinoma: a population-based analysis. *Cureus*. 2021;13(9):e18276.
 33. Nghiem P, Park SY. Less toxic, more effective treatment—A win-win for patients with Merkel cell carcinoma. *JAMA Dermatol*. 2019;155(11):1223–4.
 34. Wang AJ, McCann B, Soon WCL, De Ieso PB, Bressel M, Hui A, Chua M, Kok DL. Merkel cell carcinoma: a forty-year experience at the Peter MacCallum Cancer Centre. *BMC Cancer*. 2023;23(1):30. <https://doi.org/10.1186/s12885-022-10349-1>.
 35. Kok DL, Wang A, Xu W, Chua MST, Guminski A, Veness M, et al. The changing paradigm of managing Merkel cell carcinoma in Australia: an expert commentary. *Asia Pac J Clin Oncol*. 2020;16(6):312–9.
 36. Lodde GC, Leiter U, Gesierich A, et al. Clinical course of Merkel cell carcinoma: a DeCOG multicenter study of 1049 patients. *Eur J Cancer*. 2025;221:115406.
 37. Heath M, Jaimes N, Lemos B, Mostaghimi A, Wang LC, Peñas PF, Nghiem P. Clinical characteristics of Merkel cell carcinoma at diagnosis in 195 patients: the AEIOU features. *J Am Acad Dermatol*. 2008;58(3):375–81.
 38. Engels EA, Frisch M, Goedert JJ, Biggar RJ, Miller RW. Merkel cell carcinoma and HIV infection. *Lancet*. 2002;359(9305):497–8.
 39. Hui AC, Stillie AL, Seel M, Ainslie J. Merkel cell carcinoma: 27-year experience at the Peter MacCallum Cancer Centre. *Int J Radiat Oncol Biol Phys*. 2011;80(5):1430–5.
 40. Lee B-F, Chiu N-T, Li C-Y. Value of case-based learning in a nuclear medicine clerkship. *J Am Coll Radiol*. 2013;10(2):135–41.
 41. Hymes SR, Strom EA, Fife C. Radiation dermatitis: clinical presentation, pathophysiology, and treatment 2006. *J Am Acad Dermatol*. 2006;54(1):28–46.
 42. Ryan JL. Ionizing radiation: the good, the bad, and the ugly. *J Invest Dermatol*. 2012;132(3):985–93.

Publisher’s Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.